

Course Code	21EIE508T	Course Name	VFD AND HMI PROGRAMMING	Course Category	E	PROFESSIONAL ELECTIVE	L	T	P	C
							3	0	0	3

Pre-requisite Courses	Nil	Co-requisite Courses	Nil	Progressive Courses	Nil
Course Offering Department	Electronics and Instrumentation Engineering	Data Book / Codes / Standards	Nil		

Course Learning Rationale (CLR):	<i>The purpose of learning this course is to:</i>
CLR-1:	<i>provide knowledge on the basics of inverter and its operation modes</i>
CLR-2:	<i>know the interfacing and configuration parameters for VFD</i>
CLR-3:	<i>provide knowledge on HMI Programming</i>

Course Outcomes (CO):	At the end of this course, learners will be able to:	Programme Outcomes (PO)		
		1	2	3
CO-1:	<i>use different modes of operations for VFD</i>	1		2
CO-2:	<i>select appropriate interfacing module and parameter settings</i>	2		
CO-3:	<i>develop a user interface for a particular sequence using HMI</i>	2		2

Module-1 – Variable Frequency Drive Operations	15 Hour
<i>Basics of inverter -Inverter selection - Selection of peripheral devices - Motor types - Power and signal connections-Parameters - Different types of load patterns - Different modes of operations - PU, external& net mode, JOG operation - I/O assignment - Calibration function - PTC interface - Pulse Input, V/F control – SVC - Vector Control - PM Control - Auto tuning (offline, online), speed, torque, position control - Dynamic braking - Regenerative - braking - VFD Operational Faults and operational key based troubleshooting</i>	
Module-2 - Interfacing	12 Hour
<i>I/O option interface - Closed loop interface - Optional network interfaces- RS485, MODBUS, field bus, Ethernet, SSCNET - Inverter control using PLC - Direct communication interface with MODBUS, CC-Link, IE Communication - Troubleshooting -error massage, warning, alarm, major faults, reset function - Advance features - PID function -Brake sequence function</i>	
Module-3 – Human Machine Interface	18 Hour
<i>HMI monitor and enclosure - alarms, event history, trend, graphics library - New Project Creation - Configuration of GOT communication with PLC - OS Selection and Installation: Boot OS - Standard Monitor OS - Communication Driver - Different types of Screens - Screen Design using Various Objects like Switch/Lamps/Text and Numerical Display - Part Display and Part Routes - Comments/ Comment Groups/ Comment Display -User Alarms: Alarm configuration and Alarm Display - Objects from Library - Recipe function - Script: Project and Screen Scripts - Screen Security and Project Security configuration - Logging Functions and Time Action - Document display - Network Monitoring -Intelligent module Monitoring - Ladder Monitoring and Editor- Case study: VFD used in Closed loop for Controlling Mechanical rotatory equipment such as Oxidation Blower, Compressor, Pump and Vibrating Feeders</i>	

Learning Resources	1. Malekar, A., Malekar, A., Learn Everything about Factory Automation: Practical Lessons on PLC, HMI, VFD, Servo Programming and Machine Automation, 2021	3. F. Gardner, R., F. Gardner, R., Introduction to Plant Automation and Controls. United States: CRC Press, 2020
	2. Gurocak, H., Gurocak, H., Industrial Motion Control: Motor Selection, Drives, Controller Tuning, Applications. Germany: Wiley, 2016	4. Samuel Guccione, James McKirahan, Human Machine Interface, Concepts and Projects, Industrial Press, 2016

Learning Assessment							
	Bloom's Level of Thinking	Continuous Learning Assessment (CLA)				Summative Final Examination (40% weightage)	
		Formative CLA-1 Average of unit test (50%)		Life-Long Learning CLA-2 (10%)			
		Theory	Practice	Theory	Practice	Theory	Practice
Level 1	Remember	20%	-	20%	-	20%	-
Level 2	Understand	20%	-	20%	-	20%	-
Level 3	Apply	30%	-	30%	-	30%	-
Level 4	Analyze	30%	-	30%	-	30%	-
Level 5	Evaluate	-	-	-	-	-	-
Level 6	Create	-	-	-	-	-	-
	Total	100 %		100 %		100 %	

Course Designers		
Experts from Industry	Experts from Higher Technical Institutions	Internal Experts
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